

Literature Review Matrix – Phishing Website Detection

S.No	Reference	Problem Addressed	Methodology / Technique	Dataset	Performance (Accuracy / Key Result)	Limitations	Research Gap
1	Zara et al., IEEE Access, 2024	Improve phishing detection	ML + Ensemble + DL	UCI (~11,055)	Ensemble model ≈99% accuracy (best).	No XAI; benchmark evaluation.	Cross-dataset validation, XAI.
2	Tang & Mahmoud, IEEE Access, 2022	Real-time phishing detection	DL + browser integration	Benchmark dataset	High accuracy with browser deployment; low false alarms.	Limited interpretability .	Lightweight deployment, XAI.
3	Karim et al., IEEE Access, 2023	URL phishing detection	Hybrid ML	Public URL datasets	Hybrid ML outperformed individual ML models.	URL-only analysis.	URL+HTML+visual features.
4	Sahingoz et al., ESWA, 2019	Fast URL detection	Lexical URL + RF/SVM/DT/ANN	PhishTank, Yahoo, DMOZ	Random Forest ≈97–98% accuracy.	URL-only features.	Hybrid multimodal detection.
5	Yang et al., Sensors, 2021	DL phishing detection	CNN + RF	Public URL datasets	CNN-RF ≈98–99% accuracy.	No webpage content.	HTML + URL + XAI.
6	Remya et al., IEEE Access, 2024	Phishing URL detection	ResMLP	Public URL datasets	Higher Accuracy/F1 than baseline ML.	No explainability.	SHAP/LIME.
7	Haq et al., Applied Sciences, 2024	Automatic URL detection	1D CNN	PhishTank + UNB + Alexa	≈99.7% accuracy.	Offline evaluation.	Real-time validation.
8	Almujahid et al., PeerJ CS, 2024	Compare ML/DL	Comparative evaluation	UCI + Mendeley	CNN best performer.	Only two datasets.	Cross-dataset benchmarking.
9	Do et al., IEEE Access Survey,	Survey DL	Survey	Multiple datasets	DL generally outperforms ML.	No standard benchmark.	Standard datasets + XAI.

	2022						
10	Yang et al., Sensors, 2021	CNN-RF	CNN+RF	Public URL datasets	Higher Accuracy/F1 than ML.	URL-only.	Visual + HTML.
11	Remya et al., IEEE Access, 2025	Hybrid phishing detection	BERT+CNN+Bi LSTM	Public URL datasets	Better than standalone DL.	High computation.	Efficient transformers.
12	Barik et al., IJMLC, 2025	Optimize CNN	EGSO-CNN	Public URL datasets	Improved CNN accuracy.	Training complexity.	Efficient deployment.
13	Tian et al., JNCA Survey, 2025	Survey	Comprehensive survey	PhishTank/UCI/OpenPhish	Summarizes state-of-the-art.	No unified benchmark.	Standard evaluation.
14	Li et al., IEEE Access Review, 2024	Review	Survey	Multiple datasets	Recent methods report ~95–99% accuracy.	Limited explainability.	Cross-dataset validation.
15	Rao et al., Scientific Reports, 2025	Mobile phishing	Hybrid ensemble	Public URL datasets	≈98.9% accuracy.	Model complexity.	Lightweight deployment.

Overall Research Gaps

- Limited cross-dataset validation
- Lack of Explainable AI
- Few real-time deployment studies
- Limited evaluation on zero-day phishing
- Heavy reliance on URL-only features
- No standardized benchmarks